

Radiographic evaluation of the quality of root canal filling performed by undergraduate students at Riyadh Elm University: A retrospective study

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INTRODUCTION

Root canal treatment (RCT) involves instrumentation and obturation of the root canal system to save teeth having pulp and periapical disease. Root canal procedure requires

a high level of precision along with technical expertise for retaining and for keeping the tooth functional for the long term.

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In endodontic, the outcome of RCT is an essential step of interest from a dentist's perspective. The most critical outcome of the RCT is the periapical status of a treated tooth, as it determines whether the ultimate goal of endodontic treatment has been achieved or not.^[1] The success of RCT relies on the quality of RCT. Therefore, it is important to evaluate the quality of RCT using different mechanisms. Failure of RCT is multifactorial and could be related to the inappropriate mechanical debridement, persistence of bacteria in the root canals and apex, poor obturation quality, over and under the extension of the root canal filling materials, and coronal leakage.^[2] Despite the success rate of 86%–98%, RCT should be evaluated clinically and radiographically intermittently to rule out any shortcomings and from preventing any irreversible damage.^[3]

Radiographic assessment of the quality of RCT has been reported in the literature by different studies, indicating that radiographic assessment could be used to indicate the chances of reinfection or any fault in the procedure of RCT.^[4-8]

Reviewing RCT done by undergraduate consistently guarantees that a sufficient standard of treatment is being conveyed, as there is no base degree of information or skill-based contribution for dental undergraduate that can be esteemed worthy. Furthermore, the current undergraduate will be future dentist; therefore, it is fundamental that the highest caliber of RCT is finished, so future patient consideration would not be undermined.^[4] In 2013, a survey study was conducted to evaluate endodontic errors among undergraduate dental students in two dental schools in Riyadh, Saudi Arabia; the reported percentage of endodontic errors was 68%.^[9] A recent questionnaire-based study reported that 56% of the 469 undergraduates from two dental schools in Riyadh reported at least one endodontic procedural error during their training.^[8] The high rates of root canal fillings error indicate insufficiencies in the practical knowledge, and this information is useful in improving the dental practices and outcomes of the treatment.

Therefore, it is important to audit the quality of RCT performed by undergraduate students promptly for improving the quality of educational programs and for improving the current practice.

The aim of the present study was to assess the quality of root canal filling and the frequency of procedural errors in endodontic treatment performed by undergraduate female students at Riyadh Elm University, Riyadh, Saudi Arabia.

MATERIALS AND METHODS

This study was approved by the Research Center, College of Dentistry, Riyadh Elm University (RC/IRB/2018/1356). Data are retrieved anonymously and reported without breaching the privacy and confidentiality of the patients.

Study participants

Postendodontically treated radiograph of 306 anterior and posterior teeth done by female undergraduate dental students (4th, 5th, and 6th year) in Al Namuthajya Campus in between 2013 and 2019 were retrieved from the patient's dental record. Cases were randomly selected from the patient charts, and the following data were analyzed: (1) file number, (2) age of patient, (3) date of receiving endodontic treatment, (4) pulpal and periapical diagnosis, (5) tooth number, and (6) root canal fillings procedure.

Radiographic evaluation

To evaluate the quality of RCT of each tooth at least three digital periapical radiographs were examined using SIDEX (SIDEXIS XG, Sirona, York, Pennsylvania, United States) and Dentalplus software (Admor DentalPlus Software Ltd Phoenix Parade Artex Avenue Rustington, UK), which includes preoperative periapical radiograph, working length determination radiograph, and postoperative radiograph. All the RCTs were manually done using K-files with 0.02 taper following standard step-back technique. Irrigation was done with 2% sodium hypochlorite and saline solutions. Obturation was carried out by lateral condensation using gutta-percha and an AH Plus sealer (Dentsply Maillefer, USA).

The radiographs were evaluated independently by two senior endodontists (with minimum experience of 5 years). Each endodontist independently evaluated the selected radiograph after internal calibration, and any discordant radiologic findings were reassessed and resolved by reappraisal and discussion of the case to reach a consensus and kappa statistics was calculated for inter- and intra-examiner validity, and it was 0.89 which is considerably good.

Evaluation of root canal filling

Criteria for evaluating root canal filling quality reported by the European Society of Endodontology were followed.^[10] The criteria were as follows: acceptable RCT comprises the following characteristics: (1) uniform density from coronal to the apical aspect, (2) no voids, (3) level of root canal filling 0–2 mm from the radiographic apex, and (4) tapered. While unacceptable RCT comprises followings: (1) overfilling: root filling extending beyond the radiographic apex, (2) underfilling: root filling ending ≥ 2 mm short of

the radiographic apex, and (3) inadequate density: root filling with visible voids in the filling mass or between the filling mass and root canal walls on the final radiograph.

Exclusion criteria

Teeth with incomplete treatment, missing preoperative or postoperative periapical radiographs, poor quality radiographs, or showing <2 mm of the periapical region and radiographs that did not allow proper evaluation due to superimposition of an anatomical structure were excluded from the study.

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CONCLUSION

The quality of the root canal filling performed by undergraduate dental students was acceptable. The most frequent error was underfill of the root canal obturation. Rate of unacceptable RTCs is low with respect to other studies but reasonably higher, and therefore, auditing of RCTs done by undergraduates should be done routinely and discussed to improve the student's clinical skills for the long-term outcome.

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Conflicts of interest

There are no conflicts of interest.

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